Anh (Joe) Nguyen

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Selected Research Experience

Fujitsu Research of America — Research Intern Sep 2026 - Dec 2026

– Computer-use LLM agents: Build a self-evolving (test-time adaptation) world model for computer-use agents in enterprise environments.

Oregon State University — Research Assistant Jan 2022 - present

– Self-evolving LLM agents: Designed the contrastive sample construction that lets a self-evolving verifier learn from the agent's own trajectories, part of a training-free method for LLM agents to self-improve on open-ended tasks without any parameter updates. [2]

– LLM-based Computer use agent: Propose a simple yet effective method (RAG-PJ) to use multimodal tutorials for computer-use agents. We show that RAG-PJ achieves similar performance to other baselines *without using expensive world model rollouts*. [1]

– Language model-based RL agent: Developed an agent that can understand language descriptions of environment dynamics and act accordingly. The agent achieves *state-of-the-art policy generalization* over new language and unseen environment dynamics, increasing 6%-21% success rates over existing baselines (Dynalang, EMMA, LWM) [3]

– Visual language referring expression: Collected a new dataset in referring expressions for underwater objects. Compared and analyzed results from the state-of-the-art methods OFA and CLIP: both have low accuracy performance on underwater objects that are rare in terrestrial settings.

Singapore Management University — Research Engineer Nov 2019-Nov 2021

– Developed a method that can clarify ambiguous language queries in visual domains (referring expression): detect the most ambiguous portion of the query based on neural module network, then raise a relevant clarifying question to users.

– The method results in more efficient clarifying process: less than 10% of clarifying questions than existing baselines in CLEVR-Ref and Ref-Reasoning.

Publications

Under review (preprints)

1. Anh Nguyen, Stefan Lee. Look It Up Before Looking Ahead: Tutorials Reduce the Benefit of World Models in Computer-Use Agents, submitted to NAACL 2027.

2. Tianyi Xu, Yaolun Zhang, Yiran Wu, Anh (Joe) Nguyen, Shengyu Dai, Saba Moeinizade, Zhenwen Shao, Qingyun Wu, Huazheng Wang. Training-Free Agent Self-Improvement on Open-Ended Tasks via Evolving Feedback, submitted to ICLR 2027.

Published

3. Anh Nguyen, Stefan Lee. Language-conditioned world model improves policy generalization by reading environmental descriptions, accepted at Bridging Language, Agent, and World Models workshop at NeurIPS 2025.

4. Duy Phung, Tu Minh, Anh Nguyen, Tien Dinh, DTA Hunter System: A new statistic-based frame-

work of predicting future demand for taxi drivers, accepted for presentation @ SoICT 2017 (The Eighth International Symposium on Information and Communication Technology)

Thesis

5. Anh Nguyen, Duy Tue Tran Van, Minh Quoc Nghiem. Extractive summarization with bidirectional encoder representations from transformers. Bachelor thesis, VNU-Ho Chi Minh University of Science, 2019.

Education

2022 – 2027 (expected)	M.Sc. (thesis) and PhD at **Oregon State University**. GPA: 3.9/4.0 Advised by Professor Stefan Lee and Professor Huazheng Wang
2014 - 2019	Bachelor's Degree at **VNU-Ho Chi Minh University of Science** (GPA: 3.8/4.0, top 5% of Department) Advised by Dr. Nghiem Quoc Minh

Academic Projects

- **Language-conditioned LLM-based world model**: Identified the research gap of current LLM-based model based RL systems: they failed to incorporate language into the world model, thus unable to change world modeling on the fly based on language.
- **Exploration in RL**: Compared different exploration strategies in model-based RL: count-based, curiosity-based and Monte-Carlo dropout in PointMaze
- **Search in Games (Sudoku and (M,N,K))**: Compared different tree search in Sudoku and (M,N,K) game: Monte-Carlo Tree Search, A* and Minimax with Alpha-Beta pruning.

Awards

2025	Scholarly Presentation Award from Oregon State University.
2022	Vietnam Education Foundation (VEF) 2.0 candidate for Ph.D. programs in the USA.
2019	Fully funded exchanges: Southeast Asia Machine Learning School, KAIST, GKS ASEAN Science (Kyungwoon University), APEC Digital Innovation Forum (Taiwan)
2016	ACM-ICPC Vietnam National Competitive Programming: Third Prize
2013	National Vietnam Olympiad in Informatics: Honorable Mention

Services

- Reviewer: NeurIPS 2026, NAACL 2027, EMNLP 2026

Skills

Programming	C++ (expert), Python (expert), Java (familiar with), competitive programming
Research (expert)	LLM training: RL training